

Synthetic Fixed-Confidence Experiments for Binary Pairwise Pareto Set Identification

Experiment report generated from `Exp_Synthetic`

July 7, 2026

Abstract

This report summarizes the completed synthetic experiments for fixed-confidence binary pairwise Pareto set identification. The experiments compare VB-EGE-practical against three fixed-confidence benchmarks: UniformFocalBorda-FC, UniformPairwiseBT-MLE-FC, and UniformPairwiseBT-BordaPlugIn-FC. The main runs contain 3400 scaling replications and 2700 benchmark replications per algorithm across all experiment cells. Across all completed main settings, the empirical error rate is 0.0 for scaling and 0.0 for benchmarks; therefore the main comparison is sample complexity, measured by mean stopping time.

1 Protocol

The task is fixed-confidence identification of the strict Pareto set under binary pairwise comparison feedback. Each run stops adaptively when the algorithm can return a Pareto set certificate at the target failure probability. The default target is $\delta = 0.05$, except in the confidence-scaling sweep where δ itself is varied. The reported stopping time counts coordinate-specific pairwise comparison pulls. Lower stopping time is better when the empirical error rate is controlled.

All four methods use sample constant 2.0, threshold constant 4.0, and log constant 4.0. The BT-MLE baseline uses balanced random pair-coordinate allocation, ridge parameter 10^{-8} , fallback ridge parameter 10^{-4} , maximum absolute latent score 20, and MLE tolerance 10^{-9} in the benchmark run. Raw outputs are stored as CSV files because the local parquet engine was unavailable; this does not change the numerical results.

Metrics and legends. The main metric is mean stopping time $\bar{\tau}$. The normalized budget $\tau/(dH_B)$ uses the instance Borda hardness H_B and is used only to put settings with different dimensions and gaps on a comparable axis. Error plots show $\log_{10}(\max\{\hat{p}_{err}, 0.5/n\})$; because every main empirical error rate is zero, those curves sit at the finite-sample floor. In line plots, the legend entries are the four algorithm names. In bar plots, the x-axis labels are algorithm names and the y-axis is log-scaled mean stopping time. In pair-cell coverage plots, the vertical dashed line at coverage 1 means one sample per pair-coordinate cell on average.

Table 1: Algorithms compared in the fixed-confidence experiments.

Name	Sampling model	Fixed-confidence rule
VB-EGE-practical	Coordinate-wise Vector-Borda estimates	Adaptive accept/reject using empirical gap certificates
UniformFocalBorda-FC	Uniform coordinate-wise Borda sampling	Same confidence threshold family without adaptive elimination
UniformPairwiseBT-MLE-FC	Uniform pair-coordinate Bradley-Terry sampling with MLE	Stop when plug-in Pareto set is certified by BT confidence radius
UniformPairwiseBT-BordaPlugIn-FC	Uniform pair-coordinate sampling with Borda plug-in	Stop when Borda plug-in certificate is satisfied

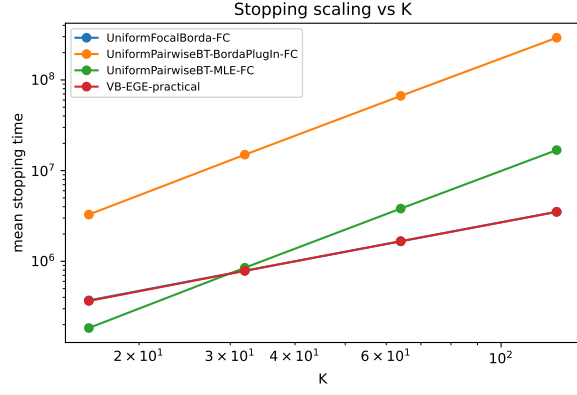
Table 2: Fixed-confidence scaling settings. All tasks use the symmetric hard generator and four algorithms; each cell is averaged over 200 random permutations/seeds.

Sweep	Generator	Grid	Purpose
K scaling	symmetric_hard	K in {16,32,64,128}; d=4; Delta=1; delta=0.05	One Pareto arm at the origin and K-1 dominated arms at -Delta in every coordinate.
d scaling	symmetric_hard	K=64; d in {2,4,10}; Delta=1; delta=0.05	Symmetric hard instance with the number of dimensions varied.
Gap scaling	symmetric_hard	K=64; d=10; Delta in {0.5,0.75,1,1.25,1.5}; delta=0.05	Symmetric hard instance with the latent separation Delta varied.
Confidence scaling	symmetric_hard	K=64; d=10; Delta=1; delta in {0.20,0.10,0.05,0.02,0.01}	Symmetric hard instance with the target failure probability varied.

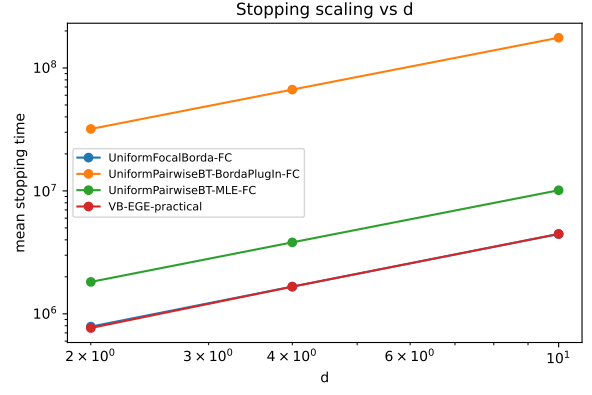
2 Fixed-Confidence Scaling Tasks

The scaling tasks isolate how stopping time changes with the number of arms, the dimension, the latent gap, and the confidence level on the symmetric hard instance. In this generator, there is a single true Pareto arm and all other arms are dominated by the same coordinate gap. This makes the setting useful for checking whether the fixed-confidence rules behave monotonically and whether pairwise baselines pay a large cost for broad pair-coordinate coverage.

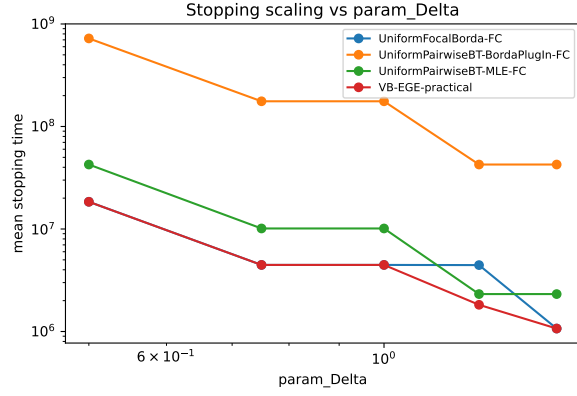
Result interpretation. VB-EGE-practical and UniformFocalBorda-FC are nearly tied on the symmetric hard instance, as expected because the instance has a single symmetric Pareto arm and little need for adaptive frontier structure. The pairwise BT-MLE and Borda plug-in baselines are substantially slower as K and d grow, reflecting the cost of spreading samples across many pair-coordinate cells. Larger gaps reduce stopping time, while smaller target delta increases stopping time through the confidence threshold. The small plateaus in the gap sweep are consistent with dyadic phase schedules: once two gap values trigger the same final phase, their stopping times can match even if the underlying gap differs.



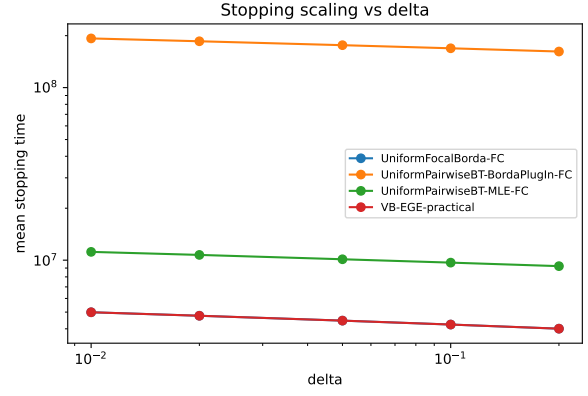
(a) K scaling



(b) d scaling



(c) Gap scaling

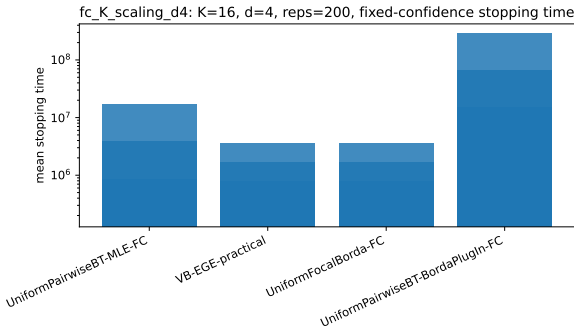


(d) Confidence scaling

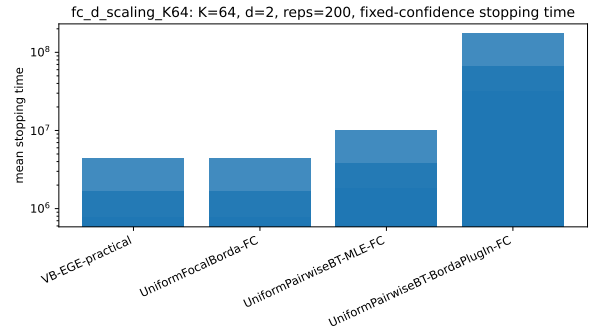
Figure 1: Scaling curves for mean stopping time. Each line is one algorithm. The axes are log-scaled; lower curves indicate fewer comparison pulls.

Table 3: Scaling results. For each swept parameter value, the table reports VB-EGE mean stopping time and baseline ratios relative to VB-EGE. All empirical error rates are zero.

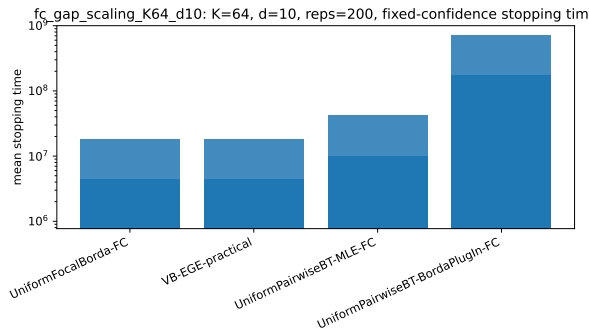
Sweep	Value	VB-EGE $\bar{\tau}$	Focal/VB	MLE/VB	Plug-in/VB
K scaling	16	3.65×10^5	1.02x	0.50x	8.97x
K scaling	32	7.84×10^5	1.00x	1.08x	19.1x
K scaling	64	1.66×10^6	1.00x	2.29x	40.1x
K scaling	128	3.51×10^6	1.00x	4.80x	83.2x
d scaling	2	7.67×10^5	1.03x	2.37x	41.6x
d scaling	4	1.66×10^6	1.00x	2.29x	40.1x
d scaling	10	4.46×10^6	1.00x	2.27x	39.5x
Gap scaling	0.50	1.84×10^7	1.00x	2.31x	39.2x
Gap scaling	0.75	4.46×10^6	1.00x	2.27x	39.5x
Gap scaling	1	4.46×10^6	1.00x	2.27x	39.5x
Gap scaling	1.25	1.82×10^6	2.44x	1.27x	23.4x
Gap scaling	1.50	1.07×10^6	1.00x	2.17x	39.8x
Confidence scaling	0.01	4.99×10^6	1.00x	2.24x	38.6x
Confidence scaling	0.02	4.76×10^6	1.00x	2.25x	39.0x
Confidence scaling	0.05	4.46×10^6	1.00x	2.27x	39.5x
Confidence scaling	0.10	4.23×10^6	1.00x	2.29x	39.9x
Confidence scaling	0.20	4.01×10^6	1.00x	2.30x	40.4x



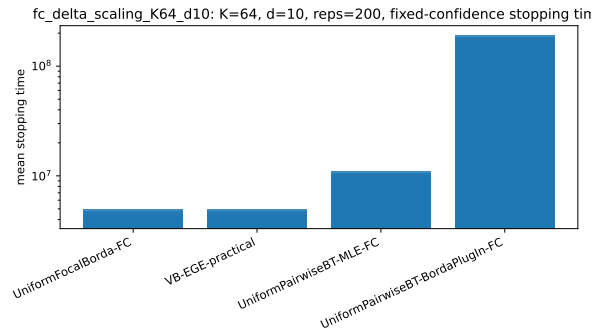
(a) K scaling



(b) d scaling

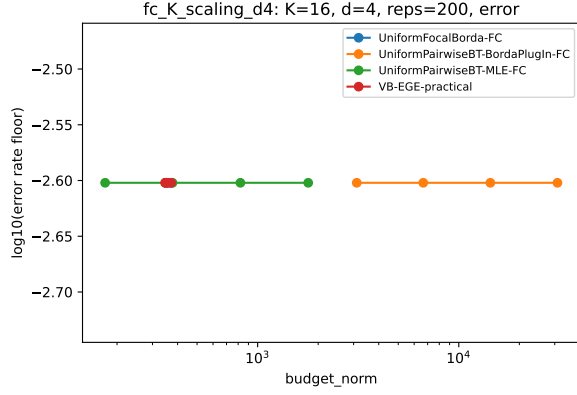


(c) Gap scaling

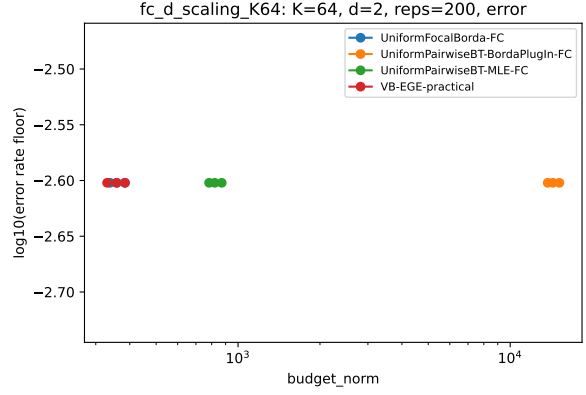


(d) Confidence scaling

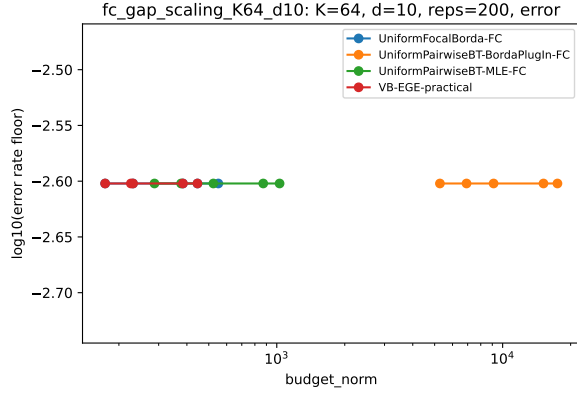
Figure 2: Per-sweep fixed-confidence stopping bars. Bars are sorted by mean stopping time within each plotted setting and use a log y-axis.



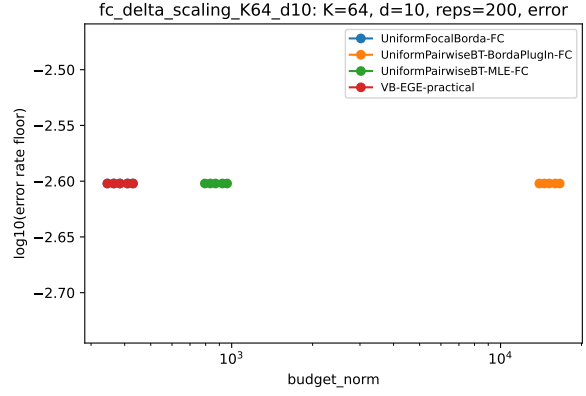
(a) K scaling



(b) d scaling



(c) Gap scaling



(d) Confidence scaling

Figure 3: Scaling error-rate floors under normalized budget. All empirical errors are zero, so the curves show the floor $0.5/n$ rather than observed failures.

Table 4: Fixed-confidence benchmark settings. All algorithms use target failure probability $\delta = 0.05$; $|P|$ denotes the true Pareto set size.

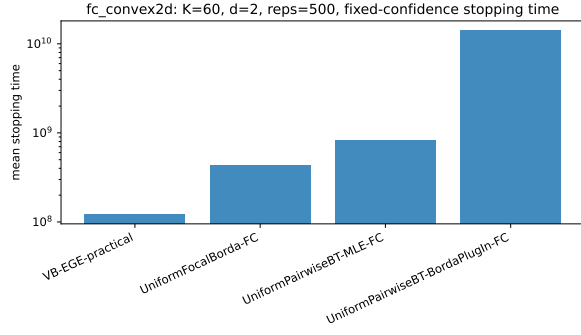
Setting	Generator	K	d	$ P $	Reps	Parameters
Convex-2D	convex_frontier_2d	60	2	15	500	s=15
Arena-4 small	arena_tradeoff_frontier	32	4	8	500	s=8, margins [0.08,0.25], alpha=0.7
Arena-4 medium	arena_tradeoff_frontier	64	4	12	300	s=12, margins [0.08,0.25], alpha=0.7
Arena-10 medium	arena_tradeoff_frontier	64	10	16	300	s=16, margins [0.06,0.20], alpha=0.7
Witness-4	unique_witness_d	40	4	8	500	s=8, q=4, margins [0.04,0.16]
Witness-10	unique_witness_d	80	10	16	300	s=16, q=4, margins [0.035,0.14]
Two-group-10	highdim_two-group	64	10	15.79	300	K_low=48, K_high=16

Table 5: Benchmark results. The table reports the mean stopping time of VB-EGE and the multiplicative stopping-time ratio of each baseline relative to VB-EGE. All empirical error rates are zero.

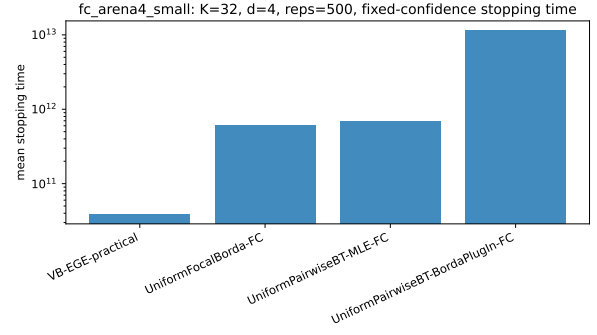
Setting	VB-EGE $\bar{\tau}$	Focal/VB	MLE/VB	Plug-in/VB	Wilson upper
Convex-2D	1.21×10^8	3.55x	6.90x	118x	0.0076
Arena-4 small	3.83×10^{10}	15.7x	18.3x	302x	0.0076
Arena-4 medium	3.65×10^{10}	31.6x	69.1x	1227x	0.0126
Arena-10 medium	9.79×10^8	13.0x	27.6x	500x	0.0126
Witness-4	2.28×10^8	1.25x	1.79x	29.7x	0.0076
Witness-10	1.61×10^9	2.15x	2.95x	48.7x	0.0126
Two-group-10	7.16×10^{11}	115x	270x	4359x	0.0126

3 Fixed-Confidence Benchmark Tasks

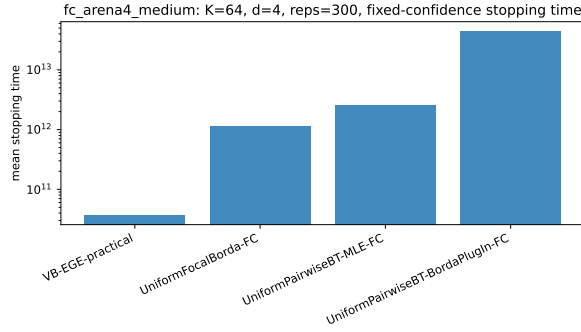
The benchmark suite stresses different Pareto geometries beyond the symmetric hard case. Convex-2D tests a clean low-dimensional trade-off frontier. Arena settings use mutually non-dominated anchors and dominated alternatives offset from a random anchor. Witness settings create dominated arms that are certified by a unique Pareto witness, which tests whether algorithms can avoid unnecessary global pairwise coverage. Two-group-10 is a high-dimensional separation task with many low-quality arms and a smaller high-quality group.



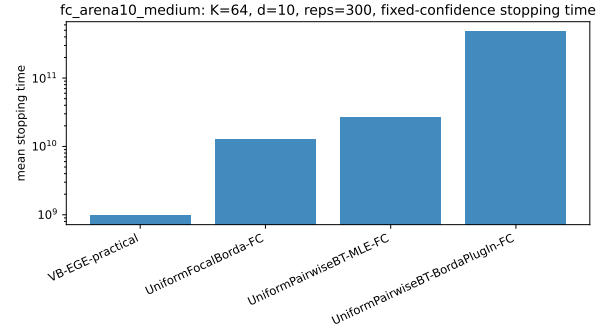
(a) Convex-2D



(b) Arena-4 small

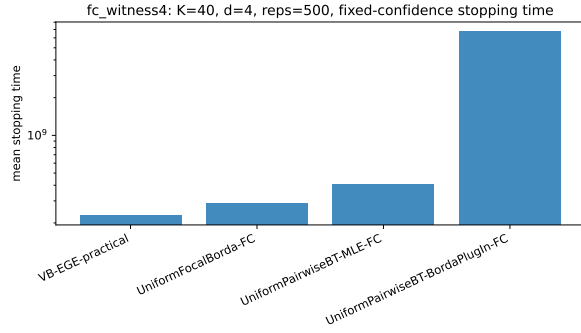


(c) Arena-4 medium

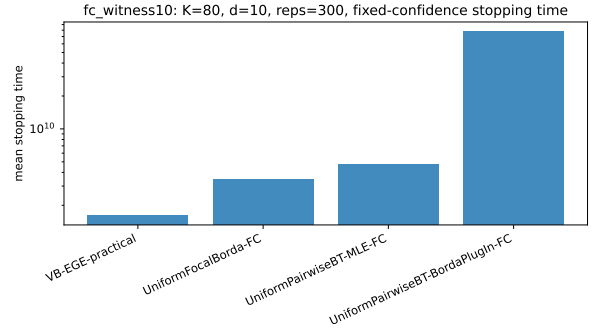


(d) Arena-10 medium

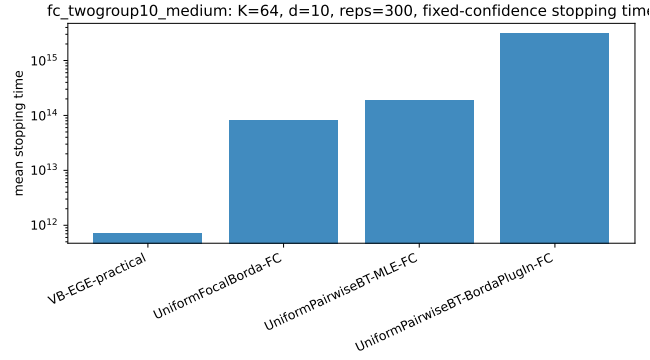
Figure 4: Benchmark stopping times, first group. Each bar is one algorithm; lower bars mean fewer comparison pulls before certification.



(a) Witness-4

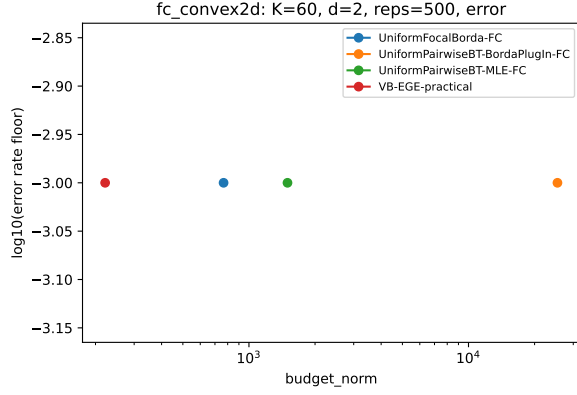


(b) Witness-10

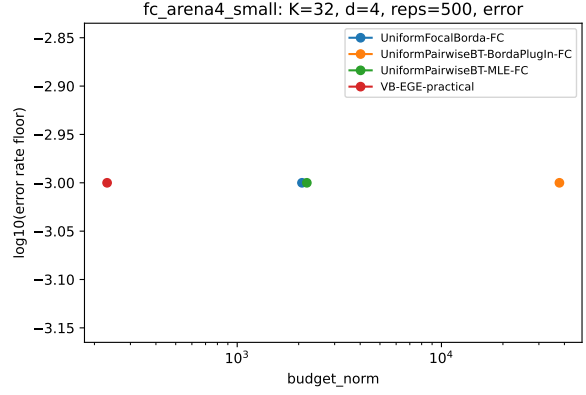


(c) Two-group-10

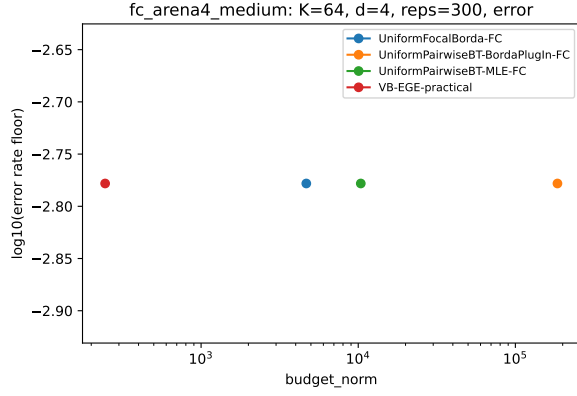
Figure 5: Benchmark stopping times, second group. These tasks emphasize local witnesses and high-dimensional group separation.



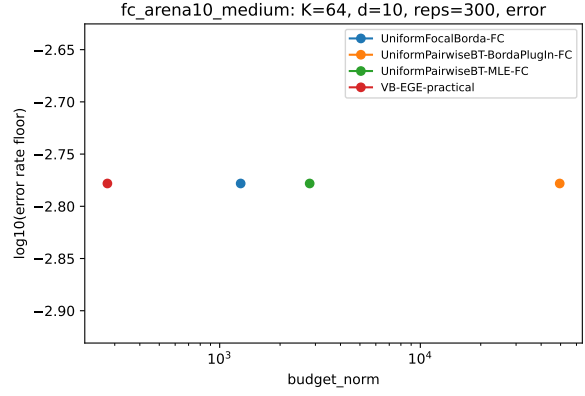
(a) Convex-2D



(b) Arena-4 small

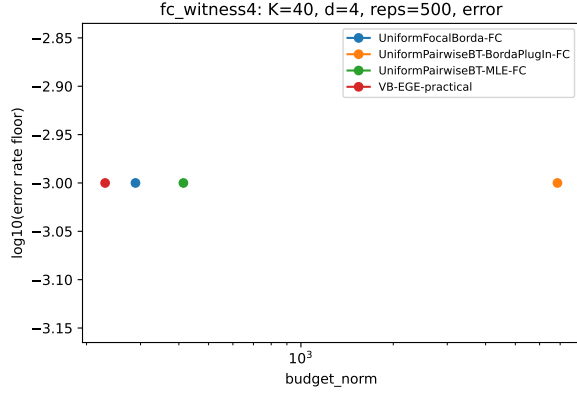


(c) Arena-4 medium

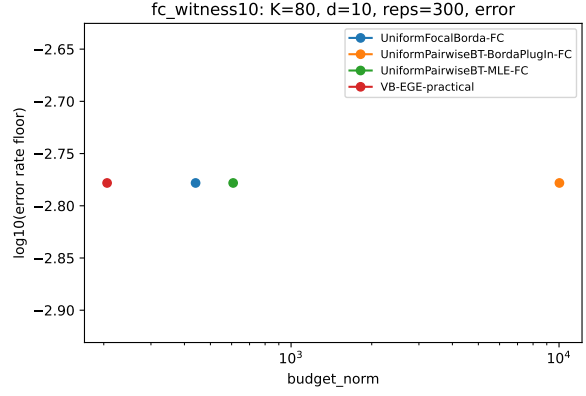


(d) Arena-10 medium

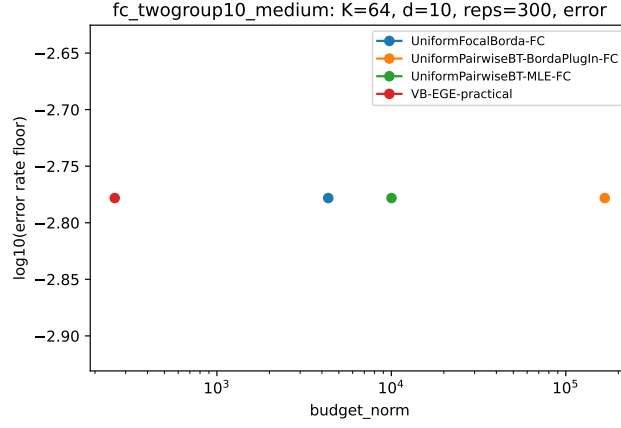
Figure 6: Benchmark normalized error floors, first group. All empirical errors are zero; the y-values are finite-sample floors.



(a) Witness-4



(b) Witness-10



(c) Two-group-10

Figure 7: Benchmark normalized error floors, second group. Curves at the same floor indicate exact empirical recovery across all replications.

Result interpretation. VB-EGE-practical has the smallest mean stopping time in every benchmark setting while maintaining zero empirical error. The advantage is moderate on unique-witness instances, where the focal Borda baseline already receives local certificate information, and much larger on arena and two-group instances. The BT-MLE baseline has near-perfect pairwise sign accuracy in the completed runs, but it pays a large exploration cost before enough pair-coordinate cells are covered. The Borda plug-in pairwise baseline is consistently the slowest, which suggests that uniform pairwise sampling is poorly matched to fixed-confidence Pareto certification when the certificate can be expressed in the Vector-Borda embedding.

4 Pair-Cell Coverage Diagnostics

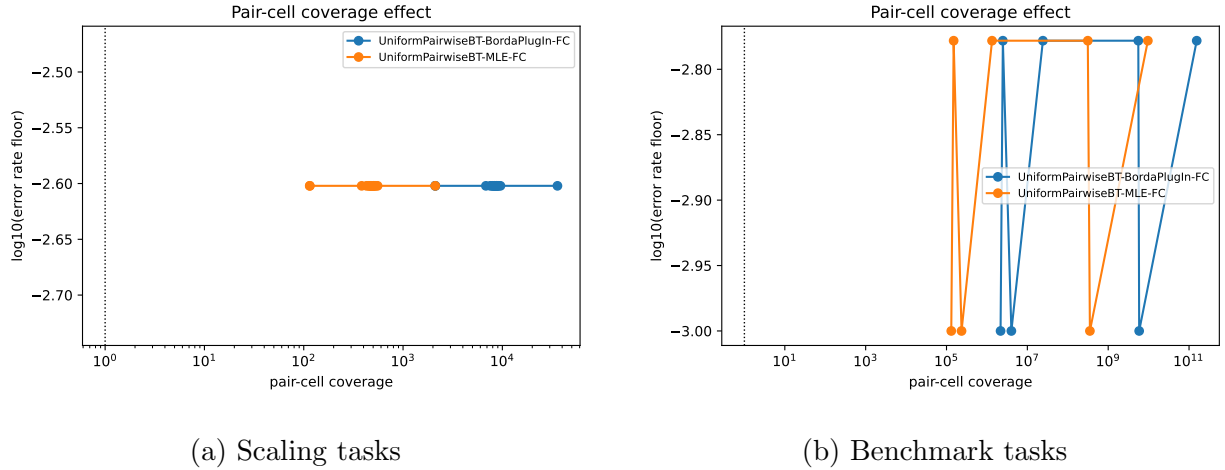


Figure 8: Pair-cell coverage diagnostics. Only pairwise baselines appear in these diagnostics. Coverage is total sampled pair-coordinate cells divided by $dK(K-1)/2$.

The coverage diagnostic explains a major source of the stopping-time gap. Pairwise baselines must collect enough direct pair-coordinate evidence to certify many signs or plug-in Borda gaps. VB-EGE instead estimates coordinate-wise Vector-Borda scores and can stop after certifying accepted and rejected arms in that lower-dimensional certificate space.

5 Takeaways

1. All completed main fixed-confidence synthetic experiments achieved zero empirical Pareto-set error.
2. VB-EGE-practical is the most sample-efficient method on every benchmark setting and is competitive with the focal Borda baseline on the symmetric hard scaling task.
3. Pairwise BT-MLE can estimate latent pairwise signs accurately, but fixed-confidence Pareto certification makes its broad pair-coordinate coverage expensive.
4. The largest gains occur in high-dimensional arena and two-group settings, where adaptive Vector-Borda certificates avoid many unnecessary pairwise comparisons.

Reproducibility Notes

The source summaries used in this report are `results/summary/fixed_confidence_scaling_summary.csv` and `results/summary/fixed_confidence_benchmarks_summary.csv`. The raw completed main outputs are `results/raw/fixed_confidence_scaling.csv` and `results/raw/fixed_confidence_benchmarks.csv`. Smoke runs are excluded from the main conclusions because they were used only to validate the implementation path before the full runs.